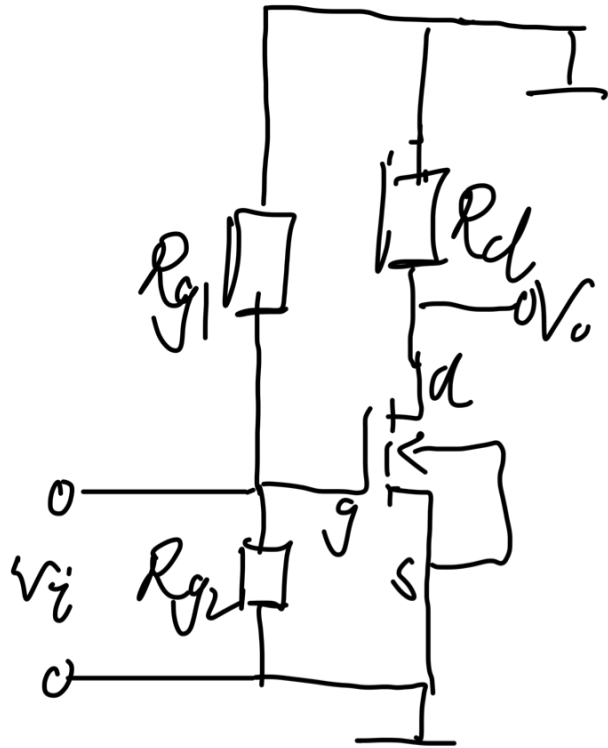
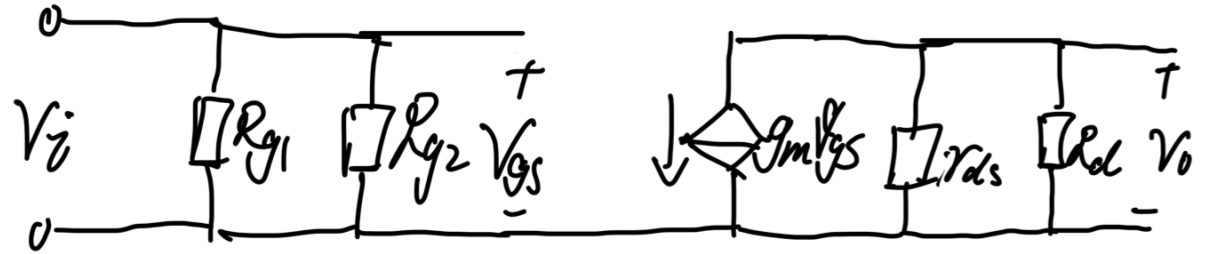




$\Rightarrow$



$$A_v = \frac{v_o}{v_i} = \frac{-g_m v_{gs} \left(\frac{r_{ds} \cdot R_d}{r_{ds} + R_d} \right)}{v_{gs}} \approx -3.1$$

$$g_m = 2K(V_{GS} - V_{th}) = 1.6 \text{ mS}$$

$$r_{ds} = \frac{1}{2K(V_{GS} - V_{th})^2} = 62.5 \text{ k}\Omega$$